22

Natural transformations

An appropriate notion of relationship between functors, giving us our first glimpse of two-dimensional structures.

We're going to develop a definition of "sensible relationships between functors", called natural transformations. We will start by a process of gut abstract feeling. This is a key example of how abstraction proceeds once you've developed some abstract intuitions — you can try more or less following your nose. This is when definitions don't need to be memorized, because they *make sense*, just like I generally hope that if I operate on the basis of respecting other humans, not hurting anyone, and being kind and helpful, then I will not break any laws, even though I'm not an expert on the law.

On the other hand, definitions of abstract structures are sometimes made by generalizing specific examples; however if you've never encountered the specific examples then that's not much motivation. Worse, if the definition *only* comes from generalizing a specific example, it has a danger of not making abstract sense, but just being somewhat utilitarian. I will present that approach in the next section, in case it helps.

22.1 Definition by abstract feeling

We start from the principle that the only possible notion of relationship between functions is equality, whereas between functors equality is too strict.

We are going to think about possible relationships between two functions f and g as shown on the right.

$$\begin{tikzcd} A \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & B \end{tikzcd}$$

Functions are defined on elements, and elements of sets do not (in general) have any notion of relationship between them except equality. Thus the only notion of relationship we can have between functions (in general) is also equality. It is defined element-wise: two functions are called equal if their action on each element produces the same element as a result.

Formally, we say $f = g$ whenever: $\boxed{\forall x \in A, \;\; f(x) = g(x).}$

Now let us move to considering relationships between two *functors* F and G as shown on the right.

$$\begin{tikzcd} \mathcal{A} \arrow[r, shift left, "F"] \arrow[r, shift right, "G"'] & \mathcal{B} \end{tikzcd}$$

Note that there is absolutely no technical reason to change the notation at this point; I'm just doing it to remind us that we've gone up a dimension. I sometimes try and write elements in lower case, sets in upper case, and categories in upper case blackboard bold or curly, but it can be tedious keeping those conventions going.

Now that we are working with categories instead of sets, we do have a notion of relationship between elements other than equality: we have morphisms.

So we can replace this form of relationship: $\forall\, x \in A, \quad f(x) = g(x)$

with this form of relationship: $\forall\, x \in \mathcal{A}, \;\; F(x) \longrightarrow G(x)$

Note that where previously we were just talking about whether or not two functions were equal, we are now looking at a system of morphisms in $\mathcal{B}$. So we are looking at "senses in which" F and G are related.

The system of morphisms we're looking at has one morphism in $\mathcal{B}$ for each object x of $\mathcal{A}$; there remains a question of what we do about morphisms in $\mathcal{A}$. We know they are mapped to morphisms in $\mathcal{B}$, so how do they relate to this system of morphisms in $\mathcal{B}$ that we've just come up with?

The action of F on morphisms gives us this: $\forall\, x \xrightarrow{f} y \in \mathcal{A}, \quad Fx \xrightarrow{Ff} Fy$

This gives us two paths from Fx to Gy: the two paths around the square shown on the right. (The horizontal morphisms are the ones giving us the relationship between F and G.)

$$\begin{tikzcd} Fx \arrow[r] \arrow[d, "Ff"'] & Gx \arrow[d, "Gf"] \\ Fy \arrow[r] & Gy \end{tikzcd}$$

In abstract structures, whenever we have two different ways of doing something, we probably want them to be related. Here we're looking at two (composite) morphisms $Fx \longrightarrow Gy$ in a category, so the only way for them to be related is by an equality. So we ask for the square to commute. This gives us the idea behind how we relate functors, as in the following definition.

Definition 22.1 Given functors $F, G\colon \mathcal{A} \longrightarrow \mathcal{B}$, a *natural transformation* α from F to G is depicted as shown on the right, or as $\alpha\colon F \Longrightarrow G$, and is given by:

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"{name=U}] \arrow[r, bend right, "G"'{name=D}] & \mathcal{B} \arrow[from=U, to=D, Rightarrow, "\alpha"] \end{tikzcd}$$

- for all objects $x \in \mathcal{A}$ a morphism $Fx \xrightarrow{\alpha_x} Gx \in \mathcal{B}$, such that
- for all morphisms $x \xrightarrow{f} y \in \mathcal{A}$ the following square called the *naturality square* commutes:

$$\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[d, "Ff"'] & Gx \arrow[d, "Gf"] \\ Fy \arrow[r, "\alpha_y"'] & Gy \end{tikzcd}$$

The morphism α_x is called the *component* of α at x.

We think of natural transformations as 2-dimensional, like a "surface" connecting the "paths" made by the functors in the diagram.

0-dimensional	categories	$\bullet$
1-dimensional	functors	$\bullet \longrightarrow \bullet$
2-dimensional	natural transformations	$\begin{tikzcd} \bullet \arrow[r, bend left] \arrow[r, bend right] \arrow[r, phantom, "\Downarrow"] & \bullet \end{tikzcd}$

Aside on the word "natural"

Eilenberg and Mac Lane observed that the point of defining categories was to define functors, and the point of defining functors is to define natural transformations.[†] The word "natural" here is both formal and evocative. It means something completely rigorous: that the naturality squares commute. But there is a certain feeling we can develop, that when some transformation feels "natural" in abstract mathematics it is a sign that there is a natural transformation at work somewhere, and, specifically, the naturality will express the real content.

22.2 Aside on homotopies

If you have seen the formal definition of homotopy then the following comparison may interest you; if not, feel free to skip it. (I am not going to *explain* the formality of homotopy here, as that is beyond our scope.)

We will start with continuous maps $f, g\colon X \longrightarrow Y$, and consider a homotopy $\alpha\colon f \Longrightarrow g$. Recall that we use the unit interval $I = [0, 1]$ and that we can think of this as a unit of "time", during which we're deforming f into g.

Then α is given by a continuous map $\boxed{\alpha\colon I \times X \longrightarrow Y}$ such that

- $\alpha(0, x) = f(x)$ ("at time 0 we are f"), and
- $\alpha(1, x) = g(x)$ ("at time 1 we are g").

We can try something similar with categories. Instead of a unit interval, we take I to be a "quintessential arrow" category, which we might write as $\boxed{0 \xrightarrow{h} 1.}$ Then, given functors $F, G\colon \mathcal{A} \longrightarrow \mathcal{B}$ we could try defining a natural transformation $\alpha\colon F \Longrightarrow G$ by analogy with the definition of homotopy. This would be a functor $\boxed{\alpha\colon I \times \mathcal{A} \longrightarrow \mathcal{B}}$ such that $\alpha(0, x) = Fx$ and $\alpha(1, x) = Gx$. In fact this gives us exactly the definition of natural transformation, just expressed rather differently.

[†] See Mac Lane, *Categories for the Working Mathematician*, Section I.4.

You might like to check that

- the components α_x are given by $F(h, 1_x)\colon (0,x) \longrightarrow (1,x)$, and
- the naturality squares are the images under F of these commutative diagrams in $I \times \mathcal{A}$:

$$\begin{tikzcd} (0,x) \arrow[r, "{(h,1_x)}"] \arrow[d, "{(1,f)}"'] & (1,x) \arrow[d, "{(1,f)}"] \\ (0,y) \arrow[r, "{(h,1_y)}"'] & (1,y) \end{tikzcd}$$

I included this for interest; we won't use it any further.

22.3 Shape

The way we have drawn natural transformations makes them 2-dimensional, with this particular shape.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \end{tikzcd}$$

This shape is a two-sided polygon, coming from the fact that F and G have the same source and target. You might wonder why they have to have the same target. The answer is that it is a choice we make because it's convenient. When we go into higher dimensions we will see that there are various choices we can make about the "shape" of higher-dimensional morphisms, because once we are in more than one dimension, things can have different shapes (whereas in 0 and 1 dimensions there's really no choice). We say that F and G have to be *parallel*, which means they have the same endpoints. The shape of α is then called *globular*.

If instead we wanted to compare *any* two functors we might look at two functors as shown on the right.

$$\begin{tikzcd} \mathcal{A} \arrow[r, "F"] & \mathcal{B} \\ \mathcal{C} \arrow[r, "G"'] & \mathcal{D} \end{tikzcd}$$

This comparison would be a bit futile without some functors going down the sides and so we'd end up needing some "vertical" functors, and then a natural transformation in the middle, as shown here.

$$\begin{tikzcd} \mathcal{A} \arrow[r, "F"] \arrow[d, "P"'] & \mathcal{B} \arrow[d, "Q"] \arrow[dl, phantom, "\alpha\,\Swarrow"] \\ \mathcal{C} \arrow[r, "G"'] & \mathcal{D} \end{tikzcd}$$

These square shapes give a different foundation for higher-dimensional thinking, called "cubical". However, in many situations we define the cubical natural transformations using the definition of globular ones; for example the one above is just a (globular) natural transformation $\alpha\colon QF \Longrightarrow GP$. So usually we might as well stick to the globular definition.

However, in some higher-dimensional situations the cubical approach turns out to be much more naturally arising, and sometimes more productive as well. We will see a glimpse of that in Chapter 24.

22.4 Functor categories

One of the principles of category theory is that if something is interesting we think about other examples of it, and morphisms between them, and assemble all that into a category. We have assembled categories into a category using functors as the morphisms, but we can now assemble functors themselves into a category, using natural transformations as the morphisms between functors.

Definition 22.2 Given categories $\mathcal{C}, \mathcal{D}$, the *functor category* $[\mathcal{C}, \mathcal{D}]$ has

- objects: functors $\mathcal{C} \longrightarrow \mathcal{D}$
- morphisms: a morphism $F \longrightarrow G$ is a natural transformation.

Identities and composition for natural transformations are componentwise.

Things To Think About

T 22.1 Can you work out the "componentwise" definition of identities and composition? Remember, "componentwise" means they operate component by component. What do we have to check to make sure this makes sense?

Definition 22.3 Given a functor $F\colon \mathcal{C} \longrightarrow \mathcal{D}$ the *identity natural transformation* $1_F\colon F \Longrightarrow F$ has all identity components.

We should just do a little type-checking: we need a component for each $x \in \mathcal{C}$, and the definition is saying that the components are $\boxed{Fx \xrightarrow{1_{Fx}} Fx.}$

The naturality squares are all trivial: for any $x \xrightarrow{f} y \in \mathcal{C}$ we have the naturality square on the right.

$$\begin{tikzcd} Fx \arrow[r, "1_{Fx}"] \arrow[d, "Ff"'] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "1_{Fy}"'] & Fy \end{tikzcd}$$

For composition there is a little more to check.

Definition 22.4 Consider functors and natural transformations as shown on the right. The composite natural transformation $\beta \circ \alpha\colon F \Longrightarrow H$ is defined by the following components: for each $x \in \mathcal{C}$, $\boxed{(\beta \circ \alpha)_x = \beta_x \circ \alpha_x.}$

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F", ""{name=U, below}] \arrow[r, "G" description, ""{name=M}] \arrow[r, bend right=60, "H"', ""{name=D, above}] & \mathcal{D} \arrow[from=U, to=M, Rightarrow, "\alpha"] \arrow[from=M, to=D, Rightarrow, "\beta"] \end{tikzcd}$$

That is: $Fx \xrightarrow{(\beta\circ\alpha)_x} Hx \quad = \quad Fx \xrightarrow{\alpha_x} Gx \xrightarrow{\beta_x} Hx.$

This is also called *vertical composition*, as it looks vertical in the 2-dimensional diagram (and there is a horizontal version which we'll see later).

We still need to check naturality: given any morphism $x \xrightarrow{f} y \in \mathcal{C}$ we need to check the square on the right.

$$\begin{tikzcd} Fx \arrow[r, "(\beta \circ \alpha)_x"] \arrow[d, "Ff"'] & Hx \arrow[d, "Hf"] \\ Fy \arrow[r, "(\beta \circ \alpha)_y"'] & Hy \end{tikzcd}$$

We can "fill this in" with the squares shown here, which commute by naturality of α and β respectively.

$$\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Hx \arrow[d, "Hf"] \\ Fy \arrow[r, "\alpha_y"'] & Gy \arrow[r, "\beta_y"'] & Hy \end{tikzcd}$$

Thus the naturality of $\beta \circ \alpha$ follows from the naturality of α and β individually.

Note that all the "action" is going on in the target category $\mathcal{D}$: the components of the natural transformation are in $\mathcal{D}$ and the naturality squares are in $\mathcal{D}$. Since composition is componentwise we inherit associativity and unit axioms from those in $\mathcal{D}$. It is a general principle that functor categories inherit more of their properties from the target category than from the source category.

Things To Think About

T 22.2 Show that we only get non-trivial natural transformations if the target category has non-trivial morphisms.

A more formal way of saying this is: if $\mathcal{D}$ is discrete (i.e. it only has identity morphisms) then the functor category $[\mathcal{C}, \mathcal{D}]$ is discrete.

This is true because any natural transformation as shown on the right must have components $\alpha_x \colon Fx \longrightarrow Gx \ \in \mathcal{D}$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$$

Thus if $\mathcal{D}$ has only identities then all the components of α must be identities. Then we must have $F = G$, and α is the identity natural transformation. This shows that the existence of this higher dimension of structure (natural transformations) comes from the morphisms in the target category. We will come back to this principle in the last chapter.

The fact that a functor category gets structure from its target category means it can be very useful to study a category $\mathcal{C}$ via functors into a category that we know has a lot of excellent structure, such as **Set**. These particular functors are so useful we give them a name, as follows.

Definition 22.5 Given a category $\mathcal{C}$, a functor $\mathcal{C}^{\text{op}} \longrightarrow \mathbf{Set}$ is called a *presheaf* on (or over) $\mathcal{C}$. The *category of presheaves over* $\mathcal{C}$ is the functor category $[\mathcal{C}^{\text{op}}, \mathbf{Set}]$.

Later we will look at why we often use contravariant functors (that is, from $\mathcal{C}^{\text{op}}$ rather than $\mathcal{C}$), and how this helps us study $\mathcal{C}$. For now we will look at a specific use of functors to express some structures we've already seen.

22.5 Diagrams and cones over diagrams

Functor categories give us a way to define cones over general diagrams, in order to define limits over general diagrams. Recall that given a small "shape" category $\mathbb{I}$, a diagram of shape $\mathbb{I}$ in a category $\mathcal{C}$ is a functor $D\colon \mathbb{I} \longrightarrow \mathcal{C}$. You can imagine $\mathbb{I}$ to be a quintessential commuting square category, or a parallel pair of arrows, if you want to visualize something. The functor D then picks out a specific diagram of that shape in the category $\mathcal{C}$.

Now, a cone over that diagram needs a vertex v and then morphisms from v to each object in the diagram, making everything in sight commute. To pick out a vertex we use a little technical trick: a constant functor. This is like a constant *function*, that sends every object to the same place. The only difference is that it's a functor so it acts on objects and morphisms: it sends every object to the same object, say v, and every morphism to the identity on v. That is the content of the following definition.

Definition 22.6 Let $\mathbb{I}$ and $\mathcal{C}$ be any categories, and v an object of $\mathcal{C}$. We write $\Delta_v\colon \mathbb{I} \longrightarrow \mathcal{C}$ for the *constant functor* at v, defined as follows.

$$\begin{array}{llcl}
 & \mathbb{I} & \xrightarrow{\Delta_v} & \mathcal{C} \\[1em]
\text{on objects:} & i & \longmapsto & v \\
\text{on morphisms:} & i \xrightarrow{f} i' & \longmapsto & v \xrightarrow{1_v} v
\end{array}$$

The idea is that no matter what the category $\mathbb{I}$ is, this functor collapses the whole thing to a single point in $\mathcal{C}$.

We now have everything we need to define a cone. This formalism might appear plucked out of thin air; we'll work out what it means immediately.

Definition 22.7 Given a diagram $D\colon \mathbb{I} \longrightarrow \mathcal{C}$, a *cone* over it with vertex v is a natural transformation $\Delta_v \Longrightarrow D$.

Things To Think About

T 22.3 Try working through this definition to understand why this gives a cone. If you're confused, you could try with some simple examples of $\mathbb{I}$, for example the quintessential arrow category.

You could also try the shapes we've used for our examples of limits: two individual objects (for products), and the shape for pullbacks as shown on the right.

$$\begin{tikzcd} & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \end{tikzcd}$$

Let's try this for the quintessential arrow category. It will help us to have names of things so let us set $\mathbb{I}$ to be this category: $\boxed{0 \xrightarrow{f} 1}$

Then a functor $D\colon \mathbb{I} \longrightarrow \mathcal{C}$ gives us this diagram in $\mathcal{C}$: $\boxed{D(0) \xrightarrow{D(f)} D(1)}$

A natural transformation $\alpha\colon \Delta_v \Longrightarrow D$ has a component $\Delta_v(i) \xrightarrow{\alpha_i} D(i)$ for each object $i \in \mathbb{I}$.

But $\Delta_v(i) = v$ for all objects, so we have components shown on the right, giving us projections from the vertex v to each object of the diagram.

$$\Delta_v(i) = v \xrightarrow{\alpha_0} D(0)$$
$$\Delta_v(i) = v \xrightarrow{\alpha_1} D(1)$$

For naturality in this case we only have one non-trivial morphism $f \in \mathbb{I}$ to consider, and its naturality square is shown on the right.

$$\begin{tikzcd} \Delta_v(0) \arrow[r, "\alpha_0"] \arrow[d, "\Delta_v(f)"'] & D(0) \arrow[d, "D(f)"] \\ \Delta_v(1) \arrow[r, "\alpha_1"'] & D(1) \end{tikzcd}$$

However, we again use the fact that Δ_v produces v for all objects and the identity for all morphisms, so the square evaluates as shown on the right.

$$\begin{tikzcd} v \arrow[r, "\alpha_0"] \arrow[d, "1_v"'] & D(0) \arrow[d, "D(f)"] \\ v \arrow[r, "\alpha_1"'] & D(1) \end{tikzcd}$$

Furthermore the identity on its left means it has effectively collapsed to a triangle as shown here. This is exactly a cone over our diagram, with vertex v.

$$\begin{tikzcd} & D(0) \arrow[dd, "D(f)"] \\ v \arrow[ur, "\alpha_0"] \arrow[dr, "\alpha_1"'] & \\ & D(1) \end{tikzcd}$$

In general, naturality corresponds to the commutativity of every triangle necessary in a cone over a diagram D, with each naturality square collapsing to one triangle. This way of expressing general cones is the basis of the general definition of "limit over a diagram of shape $\mathbb{I}$". However, the full definition is slightly beyond our scope.

22.6 Natural isomorphisms

One of the first things to do when we've made any category is to look at isomorphisms in it and see what they're like, so we'll now look at that inside functor categories. An isomorphism in a functor category is a natural transformation with an inverse; as usual the question is what that means when we unravel it.

Things To Think About

T 22.4 Can you show that isomorphisms in functor categories are componentwise isomorphisms? The first thing to do is interpret that statement.

This is saying that a natural transformation is an isomorphism iff all its components are isomorphisms. Let's state that formally.

Proposition 22.8 *A natural transformation as shown on the right is an isomorphism in the category* $[\mathcal{C}, \mathcal{D}]$ *if and only if for all* $x \in \mathcal{C}$ *the component* $\alpha_x\colon Fx \longrightarrow Gx$ *is an isomorphism in* $\mathcal{D}$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{D} \end{tikzcd}$$

Proof First suppose α has an inverse β in $[\mathcal{C}, \mathcal{D}]$, that is, a natural transformation as shown on the right such that $\beta \circ \alpha = 1_F$ and $\alpha \circ \beta = 1_G$. We aim to show that it has componentwise inverses.

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "G"] \arrow[r, bend right, "F"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{D} \end{tikzcd}$$

We just unravel this on components and we get an inverse for each component.

We claim that for each $x \in \mathcal{C}$ the component β_x is an inverse for α_x. Now by the definition of vertical composition, $(\beta \circ \alpha)_x = \beta_x \circ \alpha_x$.

So $\beta \circ \alpha = 1_F$ means: $\forall x \in \mathcal{C}$, $\beta_x \circ \alpha_x = 1_{Fx}$, as shown in the diagram on the right.

$$\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend right, dashed, "1_{Fx}"'] & Gx \arrow[r, "\beta_x"] & Fx \end{tikzcd}$$

Similarly $\alpha \circ \beta = 1_G$ means: $\forall x \in \mathcal{C}$, $\alpha_x \circ \beta_x = 1_{Gx}$, as shown on the right.

$$\begin{tikzcd} Gx \arrow[r, "\beta_x"] \arrow[rr, bend right, dashed, "1_{Gx}"'] & Fx \arrow[r, "\alpha_x"] & Gx \end{tikzcd}$$

So each β_x is an inverse for α_x, giving componentwise inverses as claimed.

For the converse we need to take individual inverses for each component of α and show that they compile into a valid natural transformation, that is, that they satisfy naturality. This part has more content.

Conversely suppose that each component α_x has an inverse β_x. We claim that the β_x are the components of a natural transformation.

We need to show that for any morphism $x \xrightarrow{f} y \in \mathcal{C}$, the square on the right commutes.

$$\begin{tikzcd} Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"'] & Fx \arrow[d, "Ff"] \\ Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}$$

The idea here, morally, is that if the top and bottom morphisms of that square are isomorphisms then the sides are "more or less the same" whichever way we travel along the isomorphisms. We need to make that rigorous though.

Consider (carefully) the diagram on the right. We are trying to show that the right-hand square commutes, and we know that all the other small regions commute, along with the outside.

$$\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}$$

Generally with commutative diagrams we know the outside commutes if all the inside parts commute. However if we are trying to show that one of the *inside* parts commutes it does not generally follow from knowing that the outside and all the other inside parts commute: care is needed.

We have the string of equalities shown below, which we read dynamically from the diagram above: at each stage we are just looking at the black arrows and ignoring the gray ones until the next step. See if you can see what commutativity is enabling us to move across each equals sign, and then see if you can read the sequence of equalities directly from the single diagram, dynamically.

$$
\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}
=
\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}
=
\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}
$$

$$
\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}
=
\begin{tikzcd} Fx \arrow[r, "\alpha_x"] \arrow[rr, bend left, "1_{Fx}"] \arrow[d, "Ff"'] & Gx \arrow[r, "\beta_x"] \arrow[d, "Gf"] & Fx \arrow[d, "Ff"] \\ Fy \arrow[r, "\alpha_y"'] \arrow[rr, bend right, "1_{Gx}"'] & Gy \arrow[r, "\beta_y"'] & Fy \end{tikzcd}
$$

This tells us that the right-hand square commutes upon pre-composition with α_x. But α_x is an isomorphism so we can cancel it out by its inverse and conclude that the right hand square commutes.

> For this last part we actually only need to know that α_x is epic or has a one-sided inverse. In algebra we have this: $Ff \circ \beta_x \circ \alpha_x = \beta_y \circ Gf \circ \alpha_x$ and so just being able to cancel α_x on one side gives us the square we want.

Thus the morphisms β_x are the components of a natural transformation that is inverse to α, as claimed. □

Definition 22.9 An invertible natural transformation is called a *natural isomorphism*. If there is a natural isomorphism $F \Longrightarrow G$ we often write $F \overset{\sim}{\Longrightarrow} G$ or $F \cong G$ and say F and G are naturally isomorphic.

We now have two equivalent characterizations of natural isomorphisms:

- abstract/categorical: isomorphisms in a functor category
- concrete/pointwise: each component is an isomorphism.

As is often the case, the abstract version is better for theory and use, where the concrete one is better for checking. That is, when we check that something is a natural isomorphism it usually comes down to checking the components, but then when we use the fact that it is a natural isomorphism we often use the fact that it is an isomorphism in a functor category, rather than the fact that its components are isomorphisms. This is often the point of having two equivalent characterizations in abstract math. One is a concrete set of conditions to check to ensure that some abstract principle holds, and the abstract principle is then the thing that enables us to do things. The key is to know how the two are related. This is often generally referred to as coherence conditions.

22.7 Equivalence of categories

The first thing we can immediately look at is how we use these "morphisms of functors" to make our more nuanced version of sameness for categories, which is called equivalence. Recall the problem with isomorphism of categories was that we invoked some equalities between objects.

This happened because isomorphism involves a pair of functors as shown on the right, such that $GF = 1$ and $FG = 1$.

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left, "F"] & \mathcal{D} \arrow[l, bend left, "G"] \end{tikzcd}$$

Now that we have the concept of isomorphism between functors we can do something more appropriately nuanced, where "appropriate" means making full use of every level of morphism available to us: the above equalities between functors are too strict, when we can use isomorphisms between functors instead. This gives us the following definition of a more nuanced notion of sameness for categories.

Definition 22.10 A functor $F\colon \mathcal{C} \longrightarrow \mathcal{D}$ is an *equivalence* if there is a functor $G\colon \mathcal{D} \longrightarrow \mathcal{C}$ and natural isomorphisms $GF \cong 1_{\mathcal{C}}$ and $FG \cong 1_{\mathcal{D}}$. Then we say the categories $\mathcal{C}$ and $\mathcal{D}$ are *equivalent*, and that F and G are *pseudo-inverses*.

Note that the prefix "pseudo" is often used in category theory when something holds up to isomorphism. Here F and G are like "inverses up to isomorphism".

There is a lot of "existence" in this definition at the moment. We characterize F by saying "there exists a pseudo-inverse" and we characterize a pseudo-inverse by saying "there exist some isomorphisms". If we wanted to be more specific we would actually ask for G and the isomorphisms to be specified. In that case it is also good to ask for a relationship between the isomorphisms. This is beyond our scope, but that structure is called an *adjoint equivalence*.

Things To Think About

T 22.5 Recall this functor F from Section 21.5, which "collapses" a fattened category to a lean one.

$$\begin{tikzcd} a \arrow[r] \arrow[dr] \arrow[d, phantom, "\cong" rotate=-90] & b \arrow[d, phantom, "\cong" rotate=-90] \\ a' \arrow[r] \arrow[ur] & b' \end{tikzcd} \quad \overset{F}{\Longrightarrow} \quad x \xrightarrow{h} y$$

Isomorphic objects are mapped to the same place, and all the non-isomorphism arrows are mapped to the same place. Show that this functor is an equivalence, by constructing a functor G going the other way, and isomorphisms $GF \cong 1_{\mathcal{C}}$ and $FG \cong 1_{\mathcal{D}}$. Is the pseudo-inverse unique?

To construct a pseudo-inverse G we need to start by deciding where x and y are going to go. This is supposed to be like an inverse, so they should be mapped to things that were in turn mapped to them; it would be silly for G to take x to b for example. However, we still have some choices: x could go to a or a', and y could go to b or b'. In either case, once we've made that decision there is no choice about where the morphism $x \longrightarrow y$ goes as there is only one non-invertible morphism available once we've fixed where x and y go; this comes down to the fact that F is full and faithful.

So we can define G as shown on the right. We then need to check the composites FG and GF.

$$\begin{array}{ccc} x & \longmapsto & a \\ y & \longmapsto & b \end{array}$$

The composite FG is actually *equal* to the identity functor since on objects the equations on the right hold, and on morphisms f is mapped back to itself.

$$\begin{array}{ccccc} FG(x) & = & F(a) & = & x \\ FG(y) & = & F(b) & = & y \end{array}$$

On the other hand GF is not *equal* to the identity functor since a' and b' are not mapped back to themselves, as shown on the right.

$$\begin{array}{ccccc} GF(a') & = & G(x) & = & a \\ GF(b') & = & G(y) & = & b \end{array}$$

However GF takes everything back to somewhere *isomorphic* to where it started, which enables us to define a natural isomorphism $GF \overset{\sim}{\Longrightarrow} 1$.

In order to do this fully it will help us to have some more names for things, so let's name everything, as shown here.

$$\mathbb{C} \qquad\qquad\qquad\qquad \mathbb{D}$$

$$\begin{tikzcd} a \arrow[r, "f_1"] \arrow[dr, "f_2" near start] \arrow[d, "p"', "\wr"] & b \arrow[d, "q", "\wr"'] \\ a' \arrow[r, "f_4"'] \arrow[ur, "f_3"' near start] & b' \end{tikzcd} \quad \overset{F}{\Longrightarrow} \quad x \xrightarrow{h} y$$

We'll now take a moment to elucidate the structure of $\mathbb{C}$. First note that $\mathbb{C}$ also has inverses to p and q which I have not drawn (and also identities, as usual). Every diagram in $\mathbb{C}$ commutes, by definition: the whole point of the morphisms

f_1, f_2, f_3, f_4 is that they are really "the same" morphism, just with the wiggle of the isomorphism built in.

So in fact in the spirit of not drawing composites that we can deduce, the category $\mathbb{C}$ could be drawn as shown here.

$$\begin{tikzcd} a \arrow[r, "f_1"] \arrow[d, "p"', "\wr"] & b \arrow[d, "q", "\wr"'] \\ a' & b' \end{tikzcd}$$

However, although it's usually clearer to omit composites, on this occasion I don't like it as it makes it look like f_1 is somehow more fundamental than, or different from, the other morphisms, when really they all play the same role.

Anyway, we now define G as shown on the right. It then remains to define a natural isomorphism $\alpha\colon GF \Longrightarrow 1_{\mathbb{C}}$.

$$\begin{array}{llcl} & \mathbb{D} & \xrightarrow{\;G\;} & \mathbb{C} \\ \text{on objects:} & x & \longmapsto & a \\ & y & \longmapsto & b \\ \text{on morphisms:} & x \xrightarrow{h} y & \longmapsto & a \xrightarrow{f_1} b \end{array}$$

> At this point if you haven't already done so you might stop and see if you can just work out where the components need to "live" (i.e. their source and target); once you've done that there's not much choice of what they actually are as morphisms.

We need one component for each object of $\mathbb{C}$, and we define them as shown in the table below.

We could think through it like this: we see what the source and target of the component are by definition, then we evaluate those objects in $\mathbb{D}$, and then we decide what morphism in $\mathbb{D}$ the component will be.

component	defined as
$\alpha_a : GFa \longrightarrow a$	$a \xrightarrow{1_a} a$
$\alpha_b : GFb \longrightarrow b$	$b \xrightarrow{1_b} b$
$\alpha_{a'} : GFa' \longrightarrow a'$	$a \xrightarrow{p} a'$
$\alpha_{b'} : GFb' \longrightarrow b'$	$b \xrightarrow{q} b'$

Each component is certainly an isomorphism, so provided this is a natural transformation it will be a natural isomorphism. Thus it just remains to check all the naturality squares. But these all live in $\mathbb{C}$, and everything in $\mathbb{C}$ commutes, so all the naturality squares must commute.

Things To Think About

T 22.6 It's still a good exercise to see if you can write down some naturality squares. There is one for each morphism of $\mathbb{C}$.

Here's the naturality square for $a \xrightarrow{f_2} b'$. First I've drawn what it is *a priori* (by definition, before we evaluate everything) and then I've drawn what it actually comes out to be in $\mathbb{C}$.

$$\begin{tikzcd} GFa \arrow[r, "\alpha_a"] \arrow[d, "GFf_2"'] & a \arrow[d, "f_2"] \\ GFb' \arrow[r, "\alpha_{b'}"'] & b' \end{tikzcd} \qquad \begin{tikzcd} a \arrow[r, "1_a"] \arrow[d, "f_1"'] & a \arrow[d, "f_2"] \\ b \arrow[r, "q"'] & b' \end{tikzcd}$$

The definition we have just been using is a "categorical" definition of equivalence, in that we have not mentioned elements of sets anywhere. In fact it's a 2-dimensional version of categorical definition: a "2-categorical" definition, as we'll see when we do higher dimensions. For now the main thing is to relate it back to the elementary definition of "pointwise equivalence" that we saw in the previous chapter. You might have felt that the above proof was a bit tedious; the elementary definition is indeed much easier to check.

Proposition 22.11 *A functor is an equivalence if and only if it is a pointwise equivalence, that is, full and faithful and essentially surjective on objects.*

This proof is a little beyond our scope. It's not hard, but it involves quite a bit of "diagram chasing" and a few little tricks, together with holding quite a lot of notation in your head at once.

See Riehl[†] for example; we have done enough of the technicalities for you to follow that proof at least step by step.

The general idea is that equivalent categories are "the same" in an appropriate sense. Categories are isomorphic if they have *exactly* the same object and arrow structure, so the only difference is that everything is renamed. Categories are equivalent if they have the same arrow structure but the objects can be "fattened up" a bit by having many isomorphic copies, or "thinned down" by identifying isomorphic objects (making them the same). In the process of fattening or thinning we just have to be careful to adjust the morphisms appropriately so that they *effectively* stay the same, as in the example above.

Things To Think About

T 22.7 Construct a two-object category $\mathbb{C}$ that is equivalent to the terminal category $\mathbb{1}$ (one object and one identity morphism). [Hint: the two objects must be uniquely isomorphic in order for $\mathbb{C}$ to be "more or less the same as" $\mathbb{1}$.] Then try doing one with three objects, and n objects.

For the example with two objects we need to take the "quintessential isomorphism" category as shown on the right; in this category $gf = 1_a$ and $fg = 1_b$.

$$\begin{tikzcd} a \arrow[r, bend left, "f"] & b \arrow[l, bend left, "g"] \end{tikzcd}$$

[†] Emily Riehl, *Category Theory in Context*, Dover, 2017.

We might draw this as shown here, to emphasize the fact that there really is no loop or hole here. The double-headed arrow indicates a sense of “reversibility”, and the symbol $\sim$ signifies isomorphism.

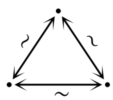
$$a \overset{\sim}{\longleftrightarrow} b$$

Now, the terminal category $\mathbb{1}$ has one object, say $*$, so a functor $\mathbb{C} \xrightarrow{F} \mathbb{1}$ must map every object to $*$. Then F being full and faithful means for every $x, y \in \mathbb{C}$ the following function on homsets is an isomorphism: $\mathbb{C}(x, y) \xrightarrow{F} \mathbb{1}(*, *)$. But the homset $\mathbb{1}(*, *)$ has only one element (the identity) so each $\mathbb{C}(x, y)$ must also have exactly one element. We'll now unravel what that means.

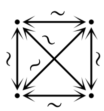
First, for any $x \in \mathbb{C}$, the homset $\mathbb{C}(x, x)$ can only contain the identity 1_x. Next, for any distinct $x, y \in \mathbb{C}$ there is exactly one morphism $x \longrightarrow y$ and one morphism $y \longrightarrow x$. Furthermore these must be inverses: the composite gf is a morphism $x \longrightarrow x$ but the only morphism $x \longrightarrow x$ is the identity 1_x. Similarly the composite $fg = 1_y$. Thus all objects of $\mathbb{C}$ must be uniquely isomorphic.

This generalizes to any category equivalent to $\mathbb{1}$, with any number of objects.

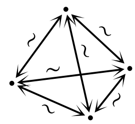
For 3 objects we could draw it like this:



For 4 objects we could draw it like this:



which we could interpret as a tetrahedron:



Categories equivalent to $\mathbb{1}$ are surprisingly useful, so we give them a name.

Definition 22.12 A category is called *indiscrete* (or *chaotic*) if it is equivalent to the terminal category.

Recall that we call a category *discrete* if it has only identity morphisms. Discrete and indiscrete categories are two ways to create a category canonically from a set of objects. Indiscrete categories might sound a bit silly as we've essentially made a bunch of objects the same from the point of view of the category, but in fact that's the whole point: it's a way to make objects the same without actually identifying them. Identifying them consists of making objects equal, which is not the right level of nuance in a category.

Indiscrete categories are used in the theory of *cliques*; I'm saying that in case you want to look them up.

Discrete and indiscrete categories have opposite universal properties. They are analogous (ideologically and formally) to discrete and indiscrete spaces in topology: in a discrete space all points are considered to be far apart, and in an indiscrete space all points are considered to be close together. The analogy is made formal by expressing the universal properties, for example by something called *adjunctions*, but this is beyond our scope; I mention it in case you would like to look it up.

22.8 Examples of equivalences of large categories

We have secretly seen various examples of equivalences of categories throughout the earlier chapters, without being able to say so. Every time we say something "is" something else in category theory, it is quite likely there is some nuance going on and that the nuance is an equivalence of categories (unless it's higher-dimensional and thus even more nuanced).

One key example is every time we have said that a monoid "is" a one-object category. In some sense this is not strictly true, as a monoid (directly defined) has as underlying data a set of elements, whereas a one-object category has an object and morphisms. But we can make sense of "is" via an equivalence of (large) categories of those two types of structure.

Things To Think About

T 22.8 See if you can construct an equivalence between the category **Mnd** of monoids and homomorphisms, and the category **ooCat** of small one-object categories and functors between them.

First note that when we take all one-object categories, we have many copies of the same monoid expressed as a one-object category in rather trivially different ways — just by having the one object being called different things.

This means that if we try and define a pair of pseudo-inverse functors as shown on the right, one direction is easier than the other.

$$\begin{tikzcd} \mathbf{Mnd} \arrow[r, bend left, "F"] & \mathbf{ooCat} \arrow[l, bend left, "G"] \end{tikzcd}$$

Defining G is "easy" as we just send a one-object category to the monoid of its morphisms. But defining F is "hard" because we have to decide *which* one-object version of the monoid to choose. However, it's not *excessively* hard: we could just decide that the single object is always going to be $*$, say. In that case GF really is the identity on **Mnd**, but FG is only isomorphic to the identity on **ooCat**. This is because if we start with a one-object category whose single object is not $*$, when we come back again via FG we will land on a category with the same monoid of morphisms, just a differently named single object. This is not *really* different, just isomorphic.

Of course, that's not a proof, but gives the idea. Incidentally the pointwise way of looking at it is that G is surjective but F is only essentially surjective.

Both are full and faithful because a functor between one-object categories is precisely a monoid homomorphism.

This is why category theorists are happy saying "a monoid is a one-object category" where others might get upset about the level shift. Those who get upset might accuse category theorists of not being rigorous, but I reckon we are being rigorous, we're just comfortable using "is" to refer to an equivalence of categories, not just an isomorphism.

22.9 Horizontal composition

I said in passing that the more abstract (not pointwise) definition of equivalence is actually 2-categorical, without saying what that means. Recall that a "categorical" definition is one that only uses objects and morphisms inside an ambient category, which means we can try taking that definition to other ambient categories to see what it gives. Our definition of equivalence didn't just use the objects and morphisms of **Cat**, as we saw that would only give us the concept of isomorphism, which is too strict for categories. Instead we escalated and used objects (categories), morphisms (functors) and "2-dimensional morphisms" (natural transformations).

The idea is that although categories and functors do form a category, there is a better, richer, more nuanced 2-dimensional environment if we take categories, functors *and* natural transformations between them. When we first defined categories, we had a prototype category given by sets and functions. Now we get a prototype 2-dimensional category given by categories, functors and natural transformations. We say "2-category" instead of "2-dimensional category" so that it's less of a mouthful.

There is one more structure that we need to consider in order to go fully 2-dimensional, and that is horizontal composition.

When we did vertical composition our natural transformations all had to be between functors going to and from the same pair of categories, as shown here.

$$\begin{tikzcd} \mathcal{C} \arrow[r, bend left=60, "F"] \arrow[r, "G" description] \arrow[r, bend right=60, "H"'] \arrow[r, phantom, shift left=3, "\Downarrow\alpha"] \arrow[r, phantom, shift right=3, "\Downarrow\beta"] & \mathcal{D} \end{tikzcd}$$

However, if we're going to organize categories, functors and natural transformations into a good structure for having them all interact, I hope you feel that this is rather restrictive. After all, we can compose functors going across the page, so what about natural transformations?

So-called horizontal composition deals with categories, functors, and natural transformations in the configuration shown here.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C} \end{tikzcd}$$

Looking at the geometry, we might expect this to produce a natural transformation as shown here.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "HF"] \arrow[r, bend right, "KG"'] \arrow[r, phantom, "\Downarrow"] & \mathcal{C} \end{tikzcd}$$

This would have components for all objects $a \in \mathcal{A}$, $\boxed{HFa \longrightarrow KGa.}$

Things To Think About

T 22.9 See if you can produce such a morphism from the components of α and β together with some functor action. In fact, there are two ways to do it; can you produce both and show that they're equal?

The components for α and β give us the morphisms shown on the right.

$$\boxed{\begin{array}{ll} \forall a \in \mathcal{A} & Fa \xrightarrow{\alpha_a} Ga \\ \forall b \in \mathcal{B} & Hb \xrightarrow{\beta_b} Kb \end{array}}$$

Now it's a case of following our nose. Starting at HFa we can make this composite.

$$\boxed{HFa \xrightarrow{H\alpha_a} HGa \xrightarrow{\beta_{Ga}} KGa.}$$

Note that for the first morphism we applied the functor H to the component α_a, but for the second one we did the component of β at the object $Ga \in \mathcal{B}$.

Here is another way to make a candidate component for the composite natural transformation we're looking for.†

$$\boxed{HFa \xrightarrow{\beta_{Fa}} KFa \xrightarrow{K\alpha_a} KGa.}$$

This time we are doing a component of β first, and a component of α afterwards. These two ways of producing a potential component $HFa \longrightarrow KGa$ look different, but are equal.

They are two ways around a naturality square for β at the morphism α_a as shown here.

$$\begin{tikzcd} HFa \arrow[r, "\beta_{Fa}"] \arrow[d, "H\alpha_a"'] & KFa \arrow[d, "K\alpha_a"] \\ HGa \arrow[r, "\beta_{Ga}"'] & KGa \end{tikzcd}$$

Now, we would like to use these morphisms $HFa \longrightarrow KFa$ (expressed either way round) to define the components of a natural transformation $HF \Longrightarrow KG$, so we need to check naturality.

Things To Think About

T 22.10 Can you check naturality for these putative components?

† That is, another way to make a morphism which goes from HFa to KGa and so has the potential to be a component of the composite natural transformation we're trying to define, which goes from HF to KG.

This is another case of filling in a diagram with smaller diagrams. Consider a morphism $x \xrightarrow{f} y \in \mathcal{A}$. I will use the first expression of the component above, that is $HFx \xrightarrow{H\alpha_x} HGx \xrightarrow{\beta_{Gx}} KGx$.

The square we need is the outside of the diagram on the right, and we fill it in as shown. We will observe that the two smaller squares are individual naturality squares.

$$\begin{tikzcd}
HFx \arrow[r, "H\alpha_x"] \arrow[d, "HFf"'] & HGx \arrow[r, "\beta_{Gx}"] \arrow[d, "HGf"] & KGx \arrow[d, "KGf"] \\
HFy \arrow[r, "H\alpha_y"'] & HGy \arrow[r, "\beta_{Gy}"'] & KGy
\end{tikzcd}$$

If you weren't able to check naturality yourself, you might take a moment now to see if you can see what naturality squares these two small ones are.

The left-hand square is the naturality square for α at the morphism f, with H applied to the whole square; recall that all functors preserve the commutativity of diagrams so after applying H the square still commutes. The right-hand square is the naturality square for β, at the morphism Gf. They both commute, so the outside commutes.

Thus we have indeed defined a natural transformation $HF \Longrightarrow KG$, and this is a new form of composition. Here is the whole definition.

Definition 22.13 Let α and β be natural transformations as shown.

$$\begin{tikzcd}
\mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C}
\end{tikzcd}$$

We define the *horizontal composite*

$$\begin{tikzcd}
\mathcal{A} \arrow[r, bend left, "HF"] \arrow[r, bend right, "KG"'] \arrow[r, phantom, "\Downarrow\beta * \alpha"] & \mathcal{C}
\end{tikzcd}$$

to have components $\quad HFa \xrightarrow{H\alpha_a} HGa \xrightarrow{\beta_{Ga}} KGa$

or equivalently: $\quad HFa \xrightarrow{\beta_{Fa}} KFa \xrightarrow{K\alpha_a} KGa.$

Note the standard notation which is to use a star $*$ for horizontal composition and a circle $\circ$ for vertical composition. Of course "horizontal" and "vertical" depend on which way up we draw things on the page; what really defines the type of composition is the dimension of the boundary along which we've stuck the natural transformations together. For vertical composition the boundary is a functor (1-dimensional) and for horizontal composition the boundary is a category (0-dimensional). When we look at higher dimensions we will see that this characterization allows us to generalize into higher dimensions.

22.10 Interchange

You might be wondering what the *meaning* is of the two different ways of defining the horizontal composite. Following our nose by type-checking can be a fun game to play, but I do like understanding the meaning of things. We are going to examine the meaning of the two ways of expressing components of the horizontal composite $\beta * \alpha$. This leads us to thinking about how horizontal composition interacts with vertical composition, which is called *interchange*.

Let's think about this composite first: $HFa \xrightarrow{H\alpha_a} HGa \xrightarrow{\beta_{Ga}} KGa.$

The first morphism $H\alpha_a$ is a component of α, with the functor H applied. We might draw it as on the right, since the functor H is applied after doing α.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, "H"] & \mathcal{C} \end{tikzcd}$$

The second morphism β_{Ga} involves applying the functor G *first* and then taking the component of β at the resulting object Ga. We might draw that as shown here.

$$\begin{tikzcd} \mathcal{A} \arrow[r, "G"'] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C} \end{tikzcd}$$

In fact those are perfectly valid ways of composing a natural transformation with a functor to produce a new natural transformation.† The first one is written $H\alpha$ and the second βG.

We can then compose $H\alpha$ and βG vertically, which we might draw like this.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, bend left, "H"] & \mathcal{C} \\ \mathcal{A} \arrow[r, bend right, "G"'] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C} \end{tikzcd}$$

This is informally referred to as *whiskering* as the functors sticking out look a bit like whiskers.

Things To Think About

T 22.11 Can you draw the diagram for whiskering the other way round, corresponding to the other way of defining horizontal composition?

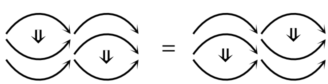
The other expression for a component of $\beta * \alpha$ is: $HFa \xrightarrow{\beta_{Fa}} KFa \xrightarrow{K\alpha_a} KGa.$

Here the first morphism says do F first and then take the component of β, so it's from the natural transformation βF. The second morphism is a component of α with K applied afterwards, so it's from the natural transformation $K\alpha$. The diagram is on the right.

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left, "F"] & \mathcal{B} \arrow[r, bend left, "H"] \arrow[r, bend right, "K"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathcal{C} \\ \mathcal{A} \arrow[r, bend left, "F"] \arrow[r, bend right, "G"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathcal{B} \arrow[r, bend right, "K"'] & \mathcal{C} \end{tikzcd}$$

We've now seen that the two expressions for a component of $\beta * \alpha$ come from two different ways of "whiskering" α and β.

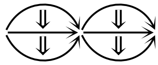
The equality of these two situations follows from the definition of naturality, and can be succinctly summed up in this diagram.



† Note that we are composing a 1-dimensional morphism (functor) with a 2-dimensional morphism (natural transformation) but it works.

For natural transformations this equality follows from the definitions, but we're going to take this as inspiration for an axiom in higher dimensions, something we demand is true in order to give a suitably coherent structure.

In fact we typically use a situation with a 2×2 grid of natural transformations as shown here.



There are two different ways of building this composite from horizontal and vertical composites:

- compose pairs horizontally first, and then compose the result vertically, or
- compose pairs vertically first, and then compose the result horizontally.

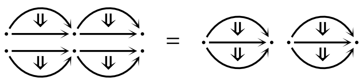
Things To Think About

T 22.12 See if you can draw diagrams to show these two different schemes, and then prove they give the same answer for natural transformations. The first thing to do is to give everything in the diagram a name.

Let us name everything in sight as follows:

$$\begin{tikzcd} \mathcal{A} \arrow[r, bend left=60, "F"] \arrow[r, "G"] \arrow[r, bend right=60, "H"'] \arrow[r, phantom, shift left=3, "\Downarrow\alpha"] \arrow[r, phantom, shift right=3, "\Downarrow\beta"] & \mathcal{B} \arrow[r, bend left=60, "P"] \arrow[r, "Q"] \arrow[r, bend right=60, "R"'] \arrow[r, phantom, shift left=3, "\Downarrow\gamma"] \arrow[r, phantom, shift right=3, "\Downarrow\delta"] & \mathcal{C} \end{tikzcd}$$

The equality we're looking for is summed up in this diagram, with the equation written in algebra below it.[†]



$$(\delta * \beta) \circ (\gamma * \alpha) \quad = \quad (\delta \circ \gamma) * (\beta \circ \alpha)$$

I definitely find the diagrammatic form more intelligible. However, in the algebra note the way that the types of composition swap places, and the inside terms also swap places, whereas the outside terms stay put. This is called *interchange*.

Definition 22.14 The above equation is called the *interchange law*.

Proposition 22.15 *Natural transformations satisfy the interchange law.*

The proof is just some fiddling around with components.

[†] Remember that the algebraic notation for composition is "backwards" from how we draw the arrows.

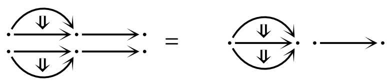
I think the hardest part is writing down what we have to show is equal, which is the outside part of the following diagram; we can then see that the square in the middle is a naturality square for γ at the morphism β_a, so the whole thing commutes.

$$\begin{tikzcd}
PFa \arrow[dr, "P\alpha_a"] & & \\
& PGa \arrow[r, "\gamma_{Ga}"] \arrow[d, "P\beta_a"'] & QGa \arrow[d, "Q\beta_a"] & \\
& PHa \arrow[r, "\gamma_{Ha}"'] & QHa \arrow[dr, "\delta_{Ha}"] & \\
& & & RHa
\end{tikzcd}$$

I do think this is the kind of thing where if you've understood what's going on it's easier to construct the diagram yourself than decipher somebody else's.

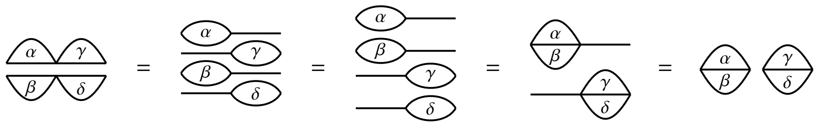
I hope that you are yearning for a more illuminating explanation; if so the following might help. Recall that we defined horizontal composition by whiskering, and started off by showing that whiskering either way round gave the same answer. We can break the interchange law down into instances of that whiskering principle.

We just need this one small thing:



If you write out what this means you will see that it follows from functoriality; we also need the version with the functor sticking out on the other side but that one follows by definition.[†]

We then have the following string of equalities.



I have omitted much of the notation here to simplify the diagrams, on the understanding that there are certain conventions: all the 1-dimensional arrows point right, all the 2-dimensional arrows point down, everything at each dimension is distinct, and so on.

Things To Think About

T 22.13 It's a good idea to verify that you know where each of these equalities is coming from.

Here is how the string of equalities arises, from left to right:

[†] This asymmetry is important in higher dimensions when we're doing things weakly, as it means things are weaker on one side than the other

1. Definition of horizontal composition.
2. Whiskering the other way round with the middle two “fish”.
3. The result above about combining the tails of two “fish”.
4. Definition of horizontal composition.

Yes, I really do think of them as fish sometimes; also the interchange law is like those little paper folding “fortune telling” things where you fold up a square of paper so that it has four corner compartments, and you stick your fingers in and make it open in opposite directions.†

Now that we have the interchange law we have everything we need to think about the totality of categories, functors, and natural transformations.

22.11 Totality

One way of approaching the definition of a category is to look at the motivating example of sets and functions and see what sorts of things they satisfy. At a basic level functions can be composed, and this composition is unital and associative. This is inherent to the definition of function, but when we define categories we then *demand* all this in axioms, and go round looking for things that satisfy it. Essentially what we’re doing is saying: “Look at this handy behavior exhibited by sets and functions. Let’s see what other places exhibit such behavior, because then we can go there and work somewhat analogously to how we work with sets and functions.”

We are now doing the same thing but instead of starting with the motivating case of sets and functions we are starting with the motivating case of categories, functors and natural transformations. This is considered to be a 2-dimensional structure, in which categories are the 0-dimensional objects, functors are 1-dimensional, and natural transformations are 2-dimensional.

There are some things we know we can do inside this world. We know that functors can be composed, and that this composition is unital and associative. We also know that natural transformations can be composed in two ways, and that those satisfy the interchange law. This is essentially the definition of a *2-category*, and we’ll do it more formally in Chapter 24 on higher dimensions. But first we will spend one chapter taking in the view from where we are.

† If you don’t know what I’m talking about you can do a search for “origami fortune teller” and it should come up.